C-Grade: Ubuntu Lab Deployment & Setup Guide

Offline C Grading System • Zero-GUI Terminal Execution Guide

Phase 1: System Packages & Hardware Profiling

Install compilers, sanitizers, Valgrind, and Linux performance profiling tools:

```
sudo apt update && sudo apt install -y \
  build-essential gcc cppcheck valgrind \
  linux-tools-common linux-tools-generic \
  python3-pip python3-venv curl git
```

Enable hardware CPU performance counter monitoring (for complexity profiling):

```
sudo sysctl -w kernel.perf_event_paranoid=1
```

Phase 2: Install & Configure Local Ollama (LLM Review)

Set up the local model daemon for code readability analysis (qwen3.5:2b):

```
# 1. Install Ollama service
curl -fsSL https://ollama.com/install.sh | sh

# 2. Start and enable system daemon
sudo systemctl enable --now ollama

# 3. Pull model weights
ollama pull qwen3.5:2b

# 4. Warm up weights into RAM
ollama run qwen3.5:2b "Say hello in one word"
```

Phase 3: Repository Setup & Python Environment

Create a native virtual environment linked to the lab machine's Python runtime:

```
# 1. Clone repository
git clone <YOUR_GITHUB_REPO_URL>
cd <YOUR_REPO_FOLDER_NAME>

# 2. Build local venv (never copy across machines)
python3 -m venv venv
source venv/bin/activate

# 3. Install direct dependencies
pip install --upgrade pip
pip install requests streamlit
```

Phase 4: Run Verification Tests in Terminal

Execute the specialist analyzers on the test submission:

```
# I/O Black-box & Auto-generation Test:
python test_tool.py --source submissions/str_len.c --analyzer io_tests --rules rules/str_rules.json

# AddressSanitizer, UBSan, and Memory Leak Check:
python test_tool.py --source submissions/str_len.c --analyzer sanitizer_run --rules rules/str_rules.json

# Edge Case & Oracle Counter-Example Verification:
python test_tool.py --source submissions/str_len.c --analyzer edge_case_runner --rules rules/str_rules.json

# Hardware Instruction & Complexity Profiler:
python test_tool.py --source submissions/str_len.c --analyzer complexity_estimator --rules rules/str_rules.json

# Local LLM Deterministic Style Review (qwen3.5:2b):
python test_tool.py --source submissions/str_len.c --analyzer llm_review --rules rules/str_rules.json
```

Phase 5: Viewing Reports & Launching UI

Inspect generated audit logs or start the web dashboard:

```
# View plain-text human summary:
cat reports/str_len.txt

# View structured JSON audit log:
cat reports/str_len.json

# Optional: Start web interface (open http://localhost:8501 in browser):
streamlit run ui.py
```

Phase 6: Lab Troubleshooting Quick Reference

Issue Encountered	Root Cause	Terminal Command Fix
bash: ./venv/bin/python: No such file	Copied venv folder from personal machine	`rm -rf venv && python3 -m venv venv && source venv/bin/activate && pip install requests streamlit`
pkill: Operation not permitted	Ollama running as root system daemon	`sudo systemctl restart ollama`
llm_review uses heuristic fallback	Ollama offline or model not loaded	`ollama run qwen3.5:2b "hi" && curl http://127.0.0.1:11434/api/tags`
perf: Permission denied	Kernel event paranoid setting active	`sudo sysctl -w kernel.perf_event_paranoid=1`
valgrind: command not found	Package omitted during apt setup	`sudo apt install -y valgrind`